



Post-Quantum Cryptography Migration

Part 3 - Operating Systems: Operating Systems & Endpoints

Compiled by the Spaced Research Ledger

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Operating systems are where post-quantum cryptography meets ordinary users. When an OS ships the new algorithms, every application on it can use them. When it turns them on by default, every user gets them without knowing.

This report covers six platform families. Section 1 covers Windows, Section 2 Apple's platforms, and Section 3 Linux. Section 4 covers Android and ChromeOS, Section 5 mainframes and the BSD systems, and Section 6 the real-time operating systems inside embedded devices.

The headline is that every major platform now ships post-quantum cryptography, and the real difference is the defaults. Apple turned quantum-secure TLS on by default across its entire 26-generation of operating systems. Windows made hybrid post-quantum TLS generally available in July 2026 but ships it disabled, leaving activation to administrators. Linux splits by distribution, with the newest releases defaulting on for SSH and TLS while older long-term-support releases need manual work.

Below the flagship platforms, the pattern repeats at different speeds. Android 17 brings post-quantum app and boot signing. IBM builds the algorithms into its newest mainframe and Power hardware. The BSDs quietly made post-quantum SSH key exchange the default. The embedded operating systems, hardest to update and longest-lived, mostly still rely on third-party libraries.

1. Windows Platform Crypto

Windows exposes cryptography to applications through a stack of Microsoft components. The SymCrypt library sits at the bottom, the CNG programming interface above it, and the Schannel TLS engine handles most software's secure connections. What lands in this stack eventually reaches most of the world's business desktops. This section covers how far each layer has come and what remains off by default.



Recent Developments

Hybrid post-quantum TLS reached production Windows in July 2026. The security updates for Windows 11 and Windows Server 2025 made ML-KEM hybrid key exchange generally available in Schannel [1], after earlier preview builds [1]. Three hybrid groups are supported, all restricted to TLS 1.3, and older TLS versions will not get them [2].

Shipping is not the same as on. The groups arrive disabled by default, and administrators must explicitly enable a post-quantum group ahead of the classical curves [2]. Enabling one protects only the session-key exchange. The organization's existing certificate hierarchy stays classical, addressed separately through ML-DSA certificate issuance [3]. Event log entries let administrators confirm a handshake actually negotiated a post-quantum group, since unsupported peers silently fall back [4].

The foundations are also being formally verified. SymCrypt released verified Rust implementations of ML-KEM and SHA-3, with machine-checked proofs, already used in Windows Insider builds [5]. The verification workflow is being extended to ML-DSA [5].

Current State

The API layer went generally available first. CNG gained ML-KEM and ML-DSA support in Windows Server 2025 and Windows 11 with the November 2025 update [6], after reaching Insider builds in May 2025 [7]. All three security levels of each algorithm are exposed [8]. Third-party hardware vendors ship key-storage providers that let Windows certificate services keep ML-DSA keys inside external HSMs [9]. Certificate issuance itself, through Active Directory Certificate Services, went GA in May 2026, with post-quantum CAs deployed fresh rather than upgraded [10].

Underneath, SymCrypt has carried the algorithms since late 2024, adding ML-KEM and XMSS in September and LMS and ML-DSA in December [11]. It now runs them in production across Azure, Microsoft 365, and Windows [12]. Microsoft commits to adding further algorithms as standards mature, without a fixed schedule [6]. Planned additions include SLH-DSA, LMS and XMSS in CNG, plus composite-certificate work tracking the IETF [7].

Unresolved Conflicts

One availability question is disputed. An industry article states SymCrypt now supports SLH-DSA alongside ML-KEM and ML-DSA [13]. Microsoft's own posts describe SLH-DSA as a planned addition in May 2025 [7] and still pending in May 2026 [12]. Microsoft's release notes for the update that ships it would settle the question.

2. Apple Platforms

Apple controls its platforms' cryptography end to end, from the corecrypto library and the Secure Enclave chip through the CryptoKit developer framework to system services like iMessage. That control lets Apple flip defaults for a billion devices at once. This section covers the 26-generation operating systems, where quantum-secure connections stopped being an option and became the default.

Recent Developments

Apple turned post-quantum protection on by default. Quantum-secure TLS is enabled for all system services and apps using the standard networking frameworks on iOS 26, iPadOS 26, macOS 26, tvOS 26, and watchOS 26 [14]. macOS 26 upgraded SSH with quantum-secure key exchange [14]. Even the iPhone-to-Watch link now runs ML-KEM-768 alongside classical key exchange [14]. Apple's network-preparation guide adds visionOS 26 to the platforms advertising hybrid key exchange in TLS [15].

Current State

Developers get the same algorithms through CryptoKit, which exposes ML-KEM at two security levels and ML-DSA at two levels across the 26-generation platforms [16]. The high-level encryption API builds on the X-Wing construction, combining ML-KEM with classical X25519 [17]. Swift Crypto mirrors the same APIs server-side, so Swift backends interoperate with Apple clients [17]. Third parties are building on it already, including post-quantum JWT support in the Vapor web framework [18]. Microsoft's .NET team independently confirmed the timing: .NET lacks post-quantum support on macOS precisely because Apple's frameworks only gained it with the 26 generation [19].

Apple also opened the foundations. In May 2026 it open-sourced its corecrypto ML-KEM and ML-DSA implementations, including hand-optimized assembly and formal verification proofs against the FIPS specifications [20].

The story started with messaging. PQ3, announced in February 2024, layered pre-standardization Kyber onto iMessage's ratchet protocol [21]. It remains scoped to iMessage, and a machine-checked security analysis of it was published at USENIX Security 2025 [22]. Neither PQ3 nor Signal's equivalent has yet moved identity authentication to a post-quantum signature. Both still authenticate with classical signatures, so the quantum protection covers message confidentiality, not the authentication layer [23].

The hardware layer is present but not fully certified. CryptoKit's ML-KEM and ML-DSA operations can run with Secure Enclave enforcement [17], and post-quantum keys can optionally be Enclave-protected [24]. In PQ3, the per-device signing key stays inside the Enclave [21]. But no completed FIPS 140-3 certificate specifically covering the Enclave's post-quantum operations could be located, even though the certification pathway exists [25].

Unresolved Conflicts

Two documentation gaps deserve flagging. Apple's own pages disagree on visionOS: the network guide includes it among quantum-secure platforms [15], while the developer security guide omits it [16]. A consistent platform list from Apple would settle it. Separately, a German preview piece claimed iOS 26.5 extends the PQ3 protocol to TLS, VPN, and SSH. That conflates PQ3, which Apple documents strictly as an iMessage protocol, with Apple's separately branded transport work, and no Apple primary source corroborates it [26].

3. Linux Distributions & Kernel Crypto

Linux is not one operating system but a family of distributions, each choosing when to ship new cryptography. Most inherit their capability from the OpenSSL and OpenSSH versions they package. The kernel itself is a separate story, with its own signing infrastructure. This section covers who ships what, whose defaults are on, and how deep the support now goes.

Recent Developments

The newest enterprise releases now prefer post-quantum by default. SUSE Linux Enterprise Server 16.0 enables post-quantum key exchange in its system-wide policies for SSH [27]. RHEL 10.2 added two FIPS-compatible post-quantum SSH key exchanges through Red Hat's own patches, hybrids that upstream OpenSSH declined to implement [28]. Debian 13 and Ubuntu 25.04 and later ship OpenSSH versions that offer ML-KEM hybrid SSH key exchange by default [29]. Alpine's current stable branch carries OpenSSL 3.5.7, giving it native ML-KEM, ML-DSA, and SLH-DSA with no add-on provider [30].

The kernel keeps extending its signing support. Linux 7.2 adds post-quantum ML-DSA signatures to the kernel's file-integrity subsystems [31]. Debian 13 also consolidated its LDAP stack onto OpenSSL, a plumbing change rather than a post-quantum feature [32].

Current State

Distribution support now splits cleanly by release age. Debian 13 ships OpenSSL 3.5 in its base [33], and guidance groups it with Alpine and RHEL-compatible 10 releases where simply updating is enough for working post-quantum TLS [34]. Ubuntu 24.04 LTS, by contrast, ships an OpenSSL too old for post-quantum work and requires building from source [35]. Ubuntu 26.04 LTS closes the gap, with hybrid SSH key exchange on by default and OpenSSL 3.5.5 [36], plus a kernel that verifies ML-DSA module signatures [37]. One practitioner assessment of that release notes nearly all software now supports quantum-resistant key exchange, while no browser yet supports quantum-resistant certificates, making key-exchange-only protection the sensible current target [38].

Red Hat's line shows the fastest maturation. RHEL 10.0 delivered ML-KEM through an add-on provider, with ML-DSA only a technology preview [39]. RHEL 10.1 made hybrid key exchange and ML-DSA certificates generally available across its TLS libraries, with the default crypto policy preferring post-quantum in TLS and SSH [40]. In FIPS mode, only the FIPS-curve hybrids function [39].

RHEL 10.2's FUTURE policy goes further and permits only hybrid post-quantum key exchange [41]. It also brings post-quantum definitions into PKCS #11 headers and composite signatures to container signing [41]. Backports are planned to carry preliminary support to RHEL 9 [42], with broader signature support following standards [42].

SUSE documents a two-track strategy: older service packs rely on the liboqs add-on architecture, while SLES 16.0 moves to native support across OpenSSL, libgcrypt, NSS, and GnuTLS [43]. Its TLS support centers on the same hybrid key exchange as everyone else [43]. SLES 16.0 itself went generally available in November 2025 [44].

The kernel's own cryptography moved in 2026. Native ML-DSA support landed in Linux 7.0, with module signing as the first consumer and X.509 and CMS paths built for wider use [45]. The work originated with a Red Hat engineer and slipped one merge window while kernel numbering rolled into 7.x [46]. Kernel documentation now lists ML-DSA alongside RSA and ECDSA for module signing [47].

One interoperability wrinkle: OpenSSL's CMS API forced the kernel's signing tool to use signed attributes for ML-DSA, pending an OpenSSL 4 fix [48]. Key encapsulation is earlier: proof-of-concept ML-KEM patches exist, deliberately unmerged until an in-kernel consumer such as NVMe authentication or WireGuard materializes [49].

4. Android & ChromeOS

Google's two consumer platforms take different routes. ChromeOS inherits most of its cryptography from the Chrome browser. Android splits the work between Conscrypt, its TLS provider, and Keystore, the hardware-backed key store that guards app signing and device credentials. This section covers a platform where signing has moved faster than connections.

Recent Developments

Android 17 is the signing milestone. It introduces APK Signature Scheme v3.2, a hybrid scheme pairing classical signatures with ML-DSA for app signing [50]. Conscrypt version 2.6.0 added support for hybrid key exchange, ML-KEM, ML-DSA, and SLH-DSA in its TLS and Java providers [51]. Apps that bundle their own cryptography, such as Signal, already provide post-quantum key establishment regardless of the platform stack, so third-party coverage is currently driven by app choices [52].

Current State

Keystore is the platform's post-quantum anchor. Android 17 natively supports ML-DSA key generation through the standard API, with keys held in the device's trusted hardware [53]. Verified Boot is integrating ML-DSA for boot-chain signatures, and remote attestation is beginning its transition [53]. Google Play will generate hybrid signing keys for new and opted-in apps during the Android 17 cycle, with a two-year rotation policy [53].

Key encapsulation in the hardware layer is explicitly later work: the secure hardware signs post-quantum today but does not yet do post-quantum key exchange [53]. A Samsung case study quantifies the cost, with ML-DSA's much larger keys straining the secure element and an optimized scheme cutting signing latency by about a third [54].

ChromeOS rides the browser. ML-KEM has been on by default since Chrome 131, protecting web traffic on ChromeOS along with everyone else [55]. An enterprise device policy exists to control it [56].

Chrome's Merkle Tree Certificates initiative for post-quantum certificates is a web-PKI effort that will reach ChromeOS when deployed [57]. The platform's verified boot, meanwhile, does not use the TPM for its core signature check, and no post-quantum update to that design was located [58]. Conscrypt's platform documentation on Android still lists no ML-KEM or ML-DSA support [59].

Unresolved Conflicts

The platform TLS timeline is murky. An aggregator claims Google deployed hybrid post-quantum TLS in Android's platform stack, but marks its own claim as inferred [52]. A directly tested Android 17 beta was found to lack post-quantum key exchange in Conscrypt entirely [52]. The same source marks Keystore's pre-Android-17 status as unverified [52]. A tested, current Android release either negotiating or failing to negotiate hybrid TLS in the platform stack would settle it.

5. Mainframe & BSD

At the two extremes of computing sit the mainframe, which processes much of the world's banking, and the BSD systems, which quietly supply cryptographic infrastructure to everything else. Both moved earlier than the noise suggests. This section covers IBM's stack and the BSD family, where post-quantum SSH became the default before most desktops had the option.

Recent Developments

IBM built the algorithms into hardware. The z17 mainframe ships with built-in quantum-safe security features [60]. IBM MQ on IBM i now supports creating keystores with post-quantum protection, though the operating system itself does not yet process them [61].

The BSDs are extending from key exchange into signatures. FreeBSD's base OpenSSH added experimental support for a composite post-quantum host-key signature scheme, off by default [62]. OpenBSD's development branch added the same experimental scheme [63]. Even the pfSense firewall distribution now includes post-quantum key exchange in its SSH daemon [64].

Current State

The mainframe stack is thorough. IBM's core cryptographic facility supports all parameter sets of ML-KEM and ML-DSA, in hardware through the Crypto Express coprocessor and in software [65]. Java applications on z/OS get the same algorithms, including hybrid key exchange, through IBM's provider [66]. The key-management product orchestrates post-quantum keys generated on the crypto cards [67], and the platform's cryptographic architectures document the algorithms with sample code [68].

On IBM's midrange, the picture is transitional. IBM i 7.6's cryptographic APIs gained new classical algorithms but no ML-KEM or ML-DSA [69]. The Power11 processor beneath it, though, ships with built-in quantum-safe cryptography [70]. Partner documentation describes it authenticating firmware and operating systems at boot with the algorithm that became ML-DSA [71]. All of this follows NIST's August 2024 finalization of the three core standards, after which each ecosystem moved on its own cadence [72].

The BSDs turned defaults on early. FreeBSD 15.0 compiled OpenSSL 3.5's post-quantum algorithms into the base system [73] and made hybrid ML-KEM SSH key exchange the default [74]. OpenBSD promoted the same default in its 7.7 release [75], then gave LibreSSL a public ML-KEM API [76].

LibreSSL 4.3.0 added the hybrid TLS keyshare, flowing to OpenBSD and to portable users on other systems [77], with a follow-up release refining it [78]. OpenBSD 7.9 ships OpenSSH 10.3 [79]. NetBSD lags: its stock

OpenSSL needs an add-on provider today [80], but its in-development 12.0 branch imports an OpenSSH new enough to default to post-quantum key exchange on release [81].

6. RTOS Platforms

Real-time operating systems run the machines around us: cars, factory controllers, medical devices, routers. They are small, long-lived, and rarely updated, which makes their cryptography both critical and slow to change. This section covers the major embedded platforms, where post-quantum support today mostly means a third-party library rather than the operating system itself.

Recent Developments

The evidence something can be done is arriving before platform support. A peer-reviewed evaluation ran every NIST-standardized parameter set of ML-KEM, ML-DSA, and SLH-DSA on a common microcontroller under FreeRTOS, without external memory [82]. A community developer separately posted a portable ML-KEM implementation with a FreeRTOS port, bounded memory use, and a tested emulated build. It is a contribution, not an official platform deliverable [83].

Current State

Platform-native support remains thin, and vendors fill the gap. FreeRTOS's official TLS integrations document no NIST-finalized post-quantum algorithms [84]. wolfSSL's post-quantum firmware TPM lists FreeRTOS, QNX, VxWorks, and other POSIX-capable systems as targets, with the support still in active development [85]. The same vendor's TLS library ships CNSA-compliant post-quantum algorithms across roughly forty embedded targets, VxWorks and Zephyr among them [86]. VxWorks's own last cryptography announcement predates the post-quantum standards by several release cycles [87].

QNX exposes cryptography through a swappable plugin architecture over OpenSSL 3.0, which is the mechanism by which newer providers could arrive without an OS rebuild [88][89]. BlackBerry says quantum-safety has been a priority since 2016 [90].

Zephyr routes cryptography through the PSA Crypto API implemented by Mbed TLS [91]. Mbed TLS's stated plan has been to wait on standardization, considering only hash-based signatures production-ready at its last review [92]. The liboqs project maintains continuing-integration coverage for Zephyr [93]. Research shows what is possible: a full post-quantum DTLS 1.3 stack on Zephyr completed verified handshakes faster than the classical baseline [94].

Unresolved Conflicts

BlackBerry's quantum messaging needs care. Its 2026 coverage of post-quantum readiness mostly describes its secure-communications and endpoint-management products, distinct from the QNX operating system, yet several articles blend both under one security narrative. Whether QNX itself has shipped concrete post-quantum integration is not established by available sources [95]. A QNX release note naming shipped algorithms would settle it.

About This Report

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